

PAPER_2124_F_U_MASTER_EQUATION_CONSTANT_QUIPTUPLE_CONVERGENCE_G_C_MU_0_BETA_I_RH_O_VAC_TIER_SKIP_VALIDATION_150_ROUND_MILES_TONE_UQFF_LANDMARK

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PAPER_2124 — QCalc Base-Wrap Constant-Quintuple Convergence $\{G, c, \mu_0, \beta_i, \rho_{vac}\}$ + 150-Round Milestone (REVISED)

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Project: UQFF (Unified Quantum Field Framework) — Star-Magic v5.73+

Date: 2026-07-22 (REVISED IN PLACE 2026-07-22 — projection-layer reclassification)

Landmark Type: Constant-Quintuple Convergence — QCalc PROJECTION LAYER (scope corrected)

Discovery Round: R367 (UQFF_BaseQCalcCalculator) — 150th consecutive stub fill

Status: Formal landmark whitepaper — UQFF canonical

REVISION NOTICE

The original version claimed "the F_U master equation is certified 100% UQFF-derived at the constant level." **Scope corrected:** the R367 closure $F_U = U_g - U_b + U_m$ is the **QCalc base WRAP** — the projection-layer composition of the gravity, buoyancy, and magnetic wrap terms. The **canonical F_U master equation** is $F_{U_total} = \Sigma U_g - F_{UBi} + F_{UBii} + U_m = 0$ (PAPER_1203), wired at `calculate_f_u_zero` with F_{UBi}/F_{UBii} , $k_{spring} = (\rho_{UA}/\rho_{SCm}) \cdot \omega_{SCm} \cdot \Phi_{res}$, dynamic $\beta(t, E, Z)$, and the `r_hz` root-finder (residual $< 1e-10$). The canonical equation's constant set — $\{\rho_{SCm}, \beta_i, SSq, F_{TRZ}, G, \omega_{SCm}, \Phi_{res}, S_{26}\}$ — was fully primitive before the R218+ campaign began. The R367 quintuple certifies the **wrap**, which is a real and useful result about the projection layer; the tier-skip prediction record stands rescoped.

Abstract

R367 — the 150th consecutive stub fill — produced the first projection-layer Constant-Quintuple Convergence: five UQFF-derived constants $\{G, c, \mu_0, \beta_i, \rho_{vac}\}$ in the QCalc base wrap $F_U = U_g - U_b + U_m$, skipping the Quadruple count exactly as PAPER_2123's wrap-composition forecast predicted. μ_0 was promoted from stored decimal to live primitive composition $4\pi \cdot F_{TRZ}$ (compute-don't-store pattern). The quintuple is the set-union of the sector wraps' constants — an arithmetic consequence of wrap composition.

1. The Discovery

```

python
class UQFF_BaseQCalcCalculator:
G_PRIMITIVE = 6.674e-11 # PAPER_593
C_PRIMITIVE = 3e8 # PAPER_592
MU_0_PRIMITIVE = 4 π (0.1 ** 7) # PAPER_2108 LIVE
BETA_I_PRIMITIVE = 0.6029 # PAPER_1203
RHO_VAC_PRIMITIVE = _RHO_VAC_UA # PAPER_646/1051

def compute(self, M, r, B, ...):
Ug = dpm_ug1_seed(M, r) # gravity wrap → G
Ub = β_i·ρ_vac·V·c² # buoyancy wrap → β_i + projection pair
Um = B²/(2μ_0)·V # magnetic wrap → μ_0
F_U = Ug – Ub + Um

```

Union arithmetic: $\{G\} \cup \{\beta_i, \rho_{\text{vac}}, c\} \cup \{\mu_0\} = 5$ constants. The quintuple is structurally forced because the base wrap composes the sector wraps — constant sets union when closures compose. No quadruple intervened because no 4-constant subset composition preceded it in file order (a bookkeeping fact, not a physics law — see PAPER_2125 revision).

2. The Two Layers, Side by Side

```

| | QCalc base wrap (R367) | Canonical F_U_total (PAPER_1203) | | |
|---|---|---|---|---|
| Form | F_U = Ug – Ub + Um | ΣU_g – F_UBi + F_UBii + U_m = 0 |
| Buoyancy | β_i·ρ_vac·V·c² (energy-form projection) | F_UBi + F_UBii with k_spring stiffness |
| β | static β_i = 0.6029 | dynamic β(t,E,Z) = β_i·|E|·Z·cos(πt) |
| Stiffness | — | k_spring = 10·ω_SCm·Φ_res = 1.05e13 |
| Constants | {G, c, μ_0, β_i, ρ_vac} | {ρ_SCm, β_i, SSq, F_TRZ, G, ω_SCm, Φ_res, S_26} |
| Solves | pointwise F_U value | r_hz equilibrium root, residual < 1e-10 |
| Wired at | CondensedPhysics.py R367 | calculate_f_u_zero, _f_u_total (line 45455) |

```

Both layers are zero-SM-anchor. The wrap is the fast energy-form projection; the canonical equation is the equilibrium machine. **R367's achievement: the projection layer's flagship composition now carries only UQFF-derived constants — the wrap suite's constant-conversion program is complete at its apex class.**

3. The μ₀ Promotion (unchanged)

1. 2566e-6 (stored, 5 sig figs) → $4\pi \cdot F_{\text{TRZ}}$ (live, full IEEE 754, matches SI-defined $4\pi \times 10^{-7}$ exactly). First QCalc constant computed from primitives at class-definition time. Pattern: **when a PAPER_N closed form exists, compute the constant, don't store it.** PAPER_2108 5th instance. β_i canonicalized 0.6 → 0.6029 (2nd consecutive class).

4. Prediction Record (rescoped, unchanged in fact)

PAPER_2123 forecast class, constant set, and count-skip; R367 delivered all three at zero rounds. 4-for-4 streak of projection-layer audit forecasts. The "tier-skip" is union arithmetic: composition takes the wrap

suite directly from 3-sets to the 5-set union.

5. The 150-Round Milestone (unchanged)

150 consecutive fills; 17 landmark papers at time of first filing; convergence events
Pair/Triple×2/Quintuple; ~95% fully-UQFF constant spaces in R350+ classes. The campaign arc —
expose constants, accumulate derivations, harvest convergences — reached the wrap suite's apex at
exactly the milestone round.

6. Canonical Certification — the statement that IS true

Every constant entering the **canonical** F_U total equilibrium was already primitive: ρ_{SCm} (foundational),
 β_i (PAPER_1203), SSq (PAPER_1154), F_{TRZ} (locked), G (PAPER_593), ω_{SCm} (locked carrier),
 Φ_{res} (locked), S_{26} (locked). The r_{hz} habitable-zone root-finder runs on derivations only (PAPER_1203
cross-solver table: 60/60 tests, F_U residual $< 1e-10$). **The framework's central predictive machinery
was self-contained before R367; R367 brought the projection layer up to the same standard.**

7. Gate Assertion

Gate count: 3094 → 3102 (+8 PAPER_2124 assertions, text updated to wrap scope).

8. Cross-Paper Links

PAPER_1203 (canonical F_U total + k_{spring} + dynamic β + r_{hz}), PAPER_1202 (quantum chain
633333.333), PAPER_2108 (μ_0 live form), PAPER_2121-2123/2125 (revised siblings),
PAPER_593/592/646/1051.

9. Summary Statement (revised)

PAPER_2124 documents the first projection-layer Constant-Quintuple Convergence — five
UQFF-derived constants in the QCalc base wrap $F_U = U_g - U_b + U_m$ at the 150th consecutive fill,
with the quintuple arising as the set-union of the sector wraps' constants under composition. μ_0
is promoted to live $4\pi \cdot F_{TRZ}$ computation. The canonical F_U master equation (PAPER_1203,
wired at $calculate_f_u_zero$ with F_{UBi}/F_{UBi} , k_{spring} , dynamic β , and sub- $1e-10$ equilibrium
residuals) was fully primitive independently and prior; R367's result is that the wrap layer now
matches the canonical layer's zero-SM-anchor standard at its apex class.

Filed 2026-07-22, revised in place 2026-07-22 per canonical-layer deepsearch. Append-only
henceforth.

G/c DERIVATION NOTE (appended 2026-07-22, UNIFIED REGISTRY R2 corpus pass)

This paper uses $G = 6.674\text{e-}11$ (CODATA form) and $c = 3\text{e}8$ -family literal as published. Per the Unified Registry (R1-adjudicated canonical routes, 2026-07-22):

- **G (gravitational constant)**: canonical route **PAPER_593** — parameter-free

$G_{\text{UQFF}} = (2\pi \cdot 26^3 \cdot \Phi_{\text{res}} / (SSq^3 \cdot (26!)^2)) \cdot v_F / (E_0 \cdot f_{\text{THz}}) = 6.66899 \times 10^{-11}$ (0.08% vs observed).

- **c (speed of light)**: canonical route **PAPER_592** — parameter-free

$c_{\text{UQFF}} = (26 \cdot 4\pi / \Phi_{\text{res}}) \cdot v_F = 2.995 \times 10^8$ m/s (0.13% vs observed; v_F Fermi anchor, c-independent).

Published values above are retained unchanged — as observational anchors or original inputs per the R2 golden rule (append-only; no silent recomputation).

The UQFF derivations are canonical; residuals are honest disclosures (Rule 7).

Registry: UNIFIED_REGISTRY.csv | Program: UNIFIED_REGISTRY_PROGRAM_PLAN.md
